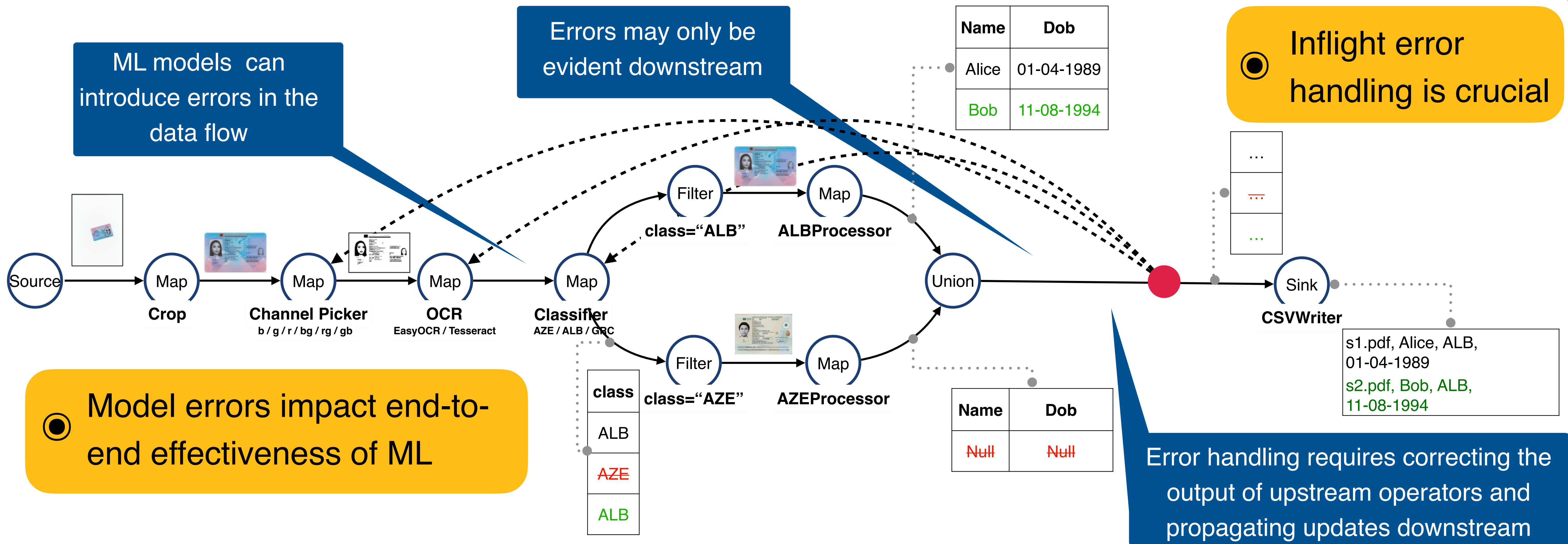


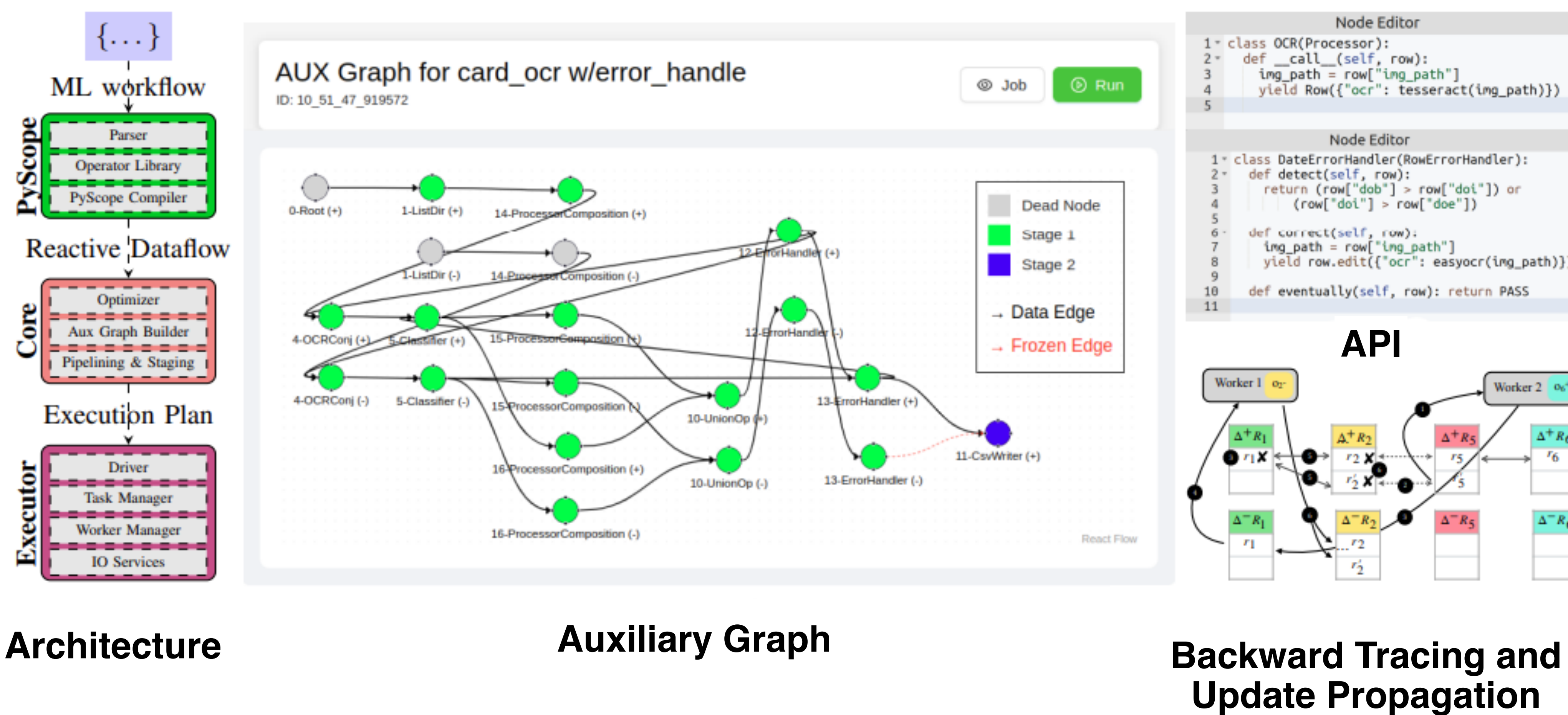
POPPER: A Dataflow System for In-Flight Error Handling in Machine Learning Workflows

Adnan Shakeel Ahmed, Abhilash Jindal, Kaustubh Beedkar

ML Workflows



Popper



✓ Dataflow system with error handlers as first-class citizens

✓ Workflows modelled as reactive dataflow

✓ Cyclic graph-based execution model

✓ Efficient pipeline parallel execution of dataflow operators

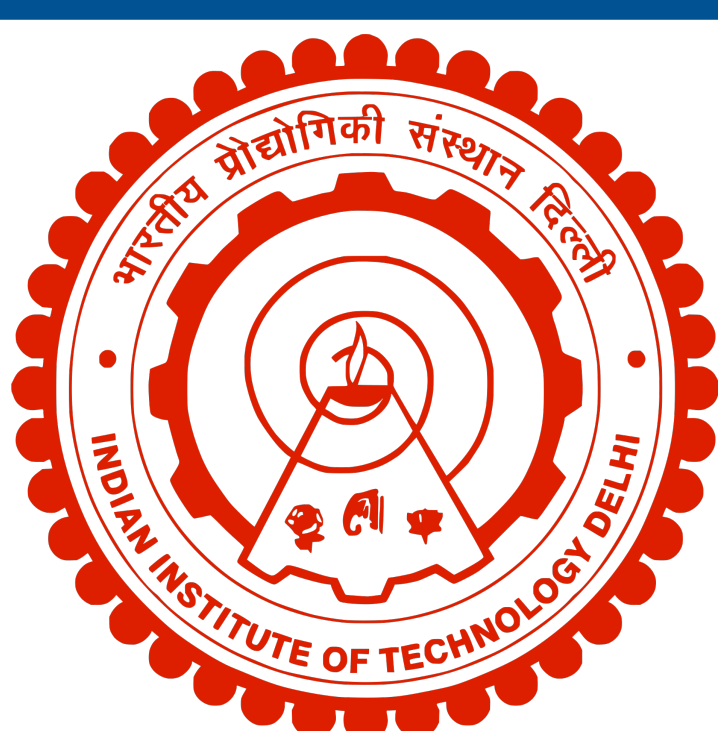
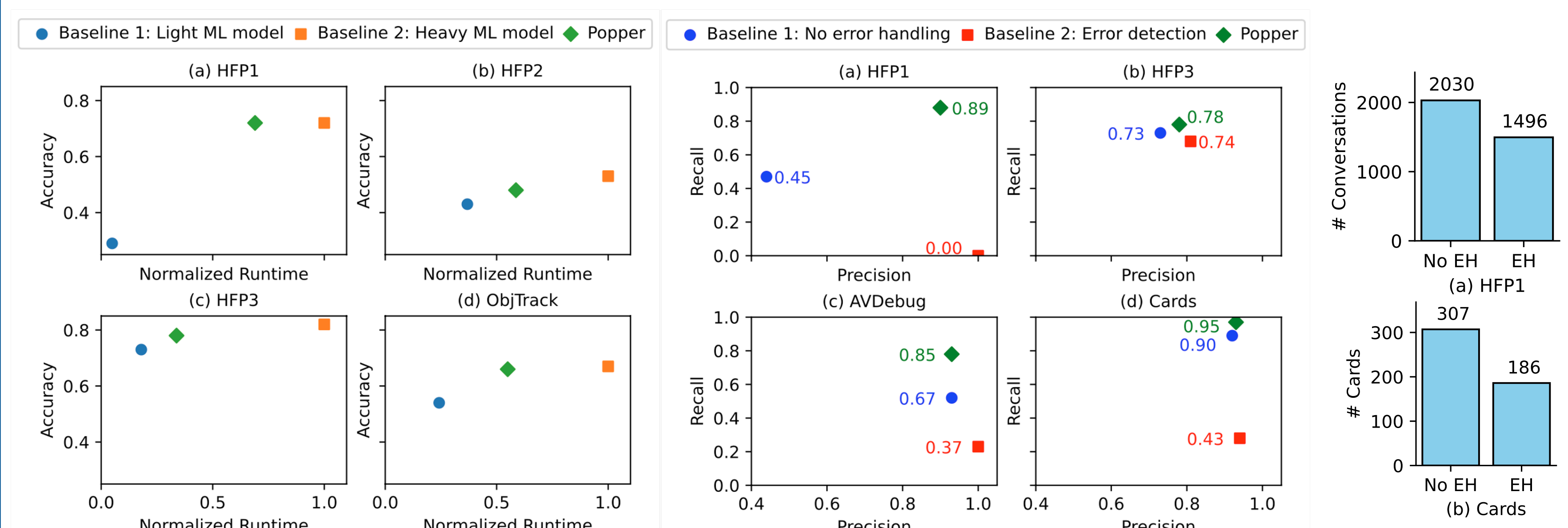
Dataflow Semantics

TABLE I: Edge labels and their semantics.

Property	$R_i + \Delta^+ R_i \rightsquigarrow$	$R_i + \Delta^- R_i \rightsquigarrow$	Label
Non-incremental	$o_j(R_i + \Delta^+ R_i)$	$o_j(R_i + \Delta^- R_i)$	$\bar{i}$
Incremental monotonic			
Only appends	$R_j + \Delta^+ R_j$	$o_j(R_i + \Delta^- R_i)$	im_+
Only deletes	$o_j(R_i + \Delta^+ R_i)$	$R_j + \Delta^- R_j$	im_-
Both	$R_j + \Delta^+ R_j$	$R_j + \Delta^- R_j$	$im_{\pm}$
Incremental non-monotonic			
Only appends	$R_j + \Delta^+ R_j + \Delta^- R_j$	$o_j(R_i + \Delta^- R_i)$	$i\bar{m}_+$
Only deletes	$o_j(R_i + \Delta^+ R_i)$	$R_j + \Delta^+ R_j + \Delta^- R_j$	$i\bar{m}_-$
Both	$R_j + \Delta^+ R_j + \Delta^- R_j$	$R_j + \Delta^+ R_j + \Delta^- R_j$	$i\bar{m}_{\pm}$

Capture transformation properties of operators with respect to handling changes in input

Evaluation



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